PH509/PHL509 Nuclear and Particle Physics
2025-26 W (Mid-Semester Exam - Part II)

Name:

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Group Name:

Common data: $m_e = 9.1 \times 10^{-31}$ kg, $m_p = m_n = 1.6 \times 10^{-27}$ kg, $h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8$ m/s, $k = 1.38 \times 10^{-23}$ J/K, $G = 6.6 \times 10^{-11}$ $\mathrm{m^3 kg^{-1} s^{-2}}$, 1 eV$= 1.6 \times 10^{-19}$ J

Each question carry 1 mark. Maximum marks will be 25. There is no negative marks. Choose the nearest order of magnitude results.

1. In Large Hadron Collider (LHC) experiments, apart from neutron n and proton p with spin $\hbar/2$, many other particles like pion π, kaon K with spin 0; ρ, K^* mesons with spin $\hbar$; Δ with spin $\frac{3\hbar}{2}$ are produced. We can classify them as
   (a) Bosons : π, K, n, p and Fermions : ρ, K^* Δ
   (b) Bosons : π, K, ρ, K^* and Fermions : n, p, Δ
   (c) Bosons : π, K, ρ, K^*, Δ and Fermions : n, p
   (d) All are Fermions
   (e) None of the above

2. Which of the following particles is a fermion?
   (a) Photon
   (b) Pion
   (c) Neutron
   (d) Gluon
   (e) None of the above

3. According to the Pauli exclusion principle,
   (a) Fermions repel each other electrically
   (b) Two identical fermions cannot occupy the same quantum state
   (c) Fermions cannot exist at low temperature
   (d) Fermions always move with relativistic speed
   (e) None of the above

4. Match the interaction with its mediating particle:
   Strong interaction $\rightarrow$?
   Electromagnetic interaction $\rightarrow$?
   Weak interaction $\rightarrow$?
   Gravitational interaction $\rightarrow$?
   (a) Photon, Gluon, W/Z bosons, Graviton
   (b) Gluon, Photon, W/Z bosons, Graviton
   (c) Gluon, W/Z bosons, Photon, Graviton
   (d) Photon, W/Z bosons, Gluon, Graviton
   (e) None of the above

5. Match Column A with the correct option from Column B:

Column A	**Column B**
A. Strong interaction	1. Acts between charged particles
B. Electromagnetic interaction	2. Responsible for radioactive beta decay
C. Weak interaction	3. Binds quarks and nucleons
D. Gravitational interaction	4. Dominant at very large masses

   (a) A–1, B–3, C–2, D–4
   (b) A–3, B–1, C–2, D–4
   (c) A–2, B–1, C–3, D–4
   (d) A–3, B–2, C–1, D–4
   (e) None of the above

6. A free neutron is unstable and undergoes beta decay, transforming into a proton while emitting an electron and an antineutrino: $n \to p + e^- + \bar{\nu}_e$. This process changes the flavour of a quark inside the neutron and is mediated by a charged intermediate vector boson. Which of the following particles is responsible for this interaction?

   (a) Photon
   (b) Gluon
   (c) $W^{\pm}$ boson
   (d) Higgs boson
   (e) None of the above

7. Fundamental interactions act selectively on particles depending on their intrinsic properties such as colour charge, electric charge, and weak isospin. Among the particles listed below, one does not carry colour charge and therefore does not experience the strong nuclear force. Which of the following particles does not participate in strong interaction?

   (a) Proton
   (b) Neutron
   (c) Electron
   (d) Pion
   (e) None of the above

8. The fact that each nucleon interacts strongly only with its nearest neighbours inside the nucleus is called:

   (a) Exchange property
   (b) Saturation property
   (c) Tensor coupling
   (d) Meson dominance
   (e) None of the above

9. In the Standard Model, elementary fermions are classified into quarks and leptons. Which of the following sets contains only leptons?

   (a) e^-, μ^-, τ^-, ν_e
   (b) u, d, s, e^-
   (c) c, s, ν_μ, μ^-
   (d) u, d, c, s

   (e) None of the above

10. Consider the following particles: u, d, e^-, ν_e, g, γ. How many of these are fermions?

    (a) 3
    (b) 4
    (c) 5
    (d) 6
    (e) None of the above

11. Which one of the following three–quark states (qqq) denoted by X cannot be a possible baryon? (The corresponding electric charge is indicated in the superscript.)

    (a) X^{++}
    (b) X^{+}
    (c) X^{-}
    (d) X^{--}
    (e) None of the above

12. The atomic masses of ${}^{152}_{63}\mathrm{Eu}$, ${}^{152}_{62}\mathrm{Sm}$, ${}^{1}_{1}\mathrm{H}$ and neutron are 151.921749, 151.919756, 1.007825 and 1.008665 amu, respectively. The Q-value is the net energy released or absorbed during a nuclear reaction. Using this information, the Q-value of the reaction ${}^{152}_{63}\mathrm{Eu} + n \to {}^{152}_{62}\mathrm{Sm} + p$ when expressed in atomic mass units (amu) is:

    (a) 3.833×10^{-3} amu
    (b) 2.833×10^{-3} amu
    (c) 3.833×10^{-4} amu

(d) 2.833×10^{-4} amu
(e) None of the above

13. In white dwarfs, the inward gravitational pressure is balanced by the outward electron degeneracy pressure. Determine the radius of such a white dwarf that has electron degeneracy pressure of 2×10^{19} Pa if it has a mass of 2.5×10^{30} kg. The expression for gravitational pressure is $P = \frac{3G}{16\pi}\frac{M^2}{R^4}$.
(a) 6.33×10^{5} m
(b) 3.33×10^{6} m
(c) 3.33×10^{7} m
(d) 6.33×10^{8} m
(e) None of the above

14. For a non-relativistic degenerate electron gas, the degeneracy pressure is given by $P = \frac{1}{5m_e}\left(\frac{3h^3}{8\pi}\right)^{2/3} n^{5/3}$, where m_e is the electron mass and n is the number density. Estimate the electron degeneracy pressure for a white dwarf where $n = 10^{30}\,\mathrm{m}^{-3}$. The pressure is of the order of
(a) 2.34×10^{12} Pa
(b) 2.34×10^{15} Pa
(c) 2.34×10^{18} Pa
(d) 2.34×10^{21} Pa
(e) None of the above

15. For a non-relativistic degenerate fermi gas, the degeneracy pressure is given by $P = \frac{1}{5m}\left(\frac{3h^3}{4\pi g}\right)^{2/3} n^{5/3}$, where m is the mass and n is the number density and g is the degenerarcy of the particle. Estimate the degeneracy pressure for a neutron star composed solely of neutrons with degeneracy of 2 where $n = 10^{30}\,\mathrm{m}^{-3}$. The pressure is of the order of
(a) 1.3×10^{13} Pa
(b) 1.3×10^{9} Pa
(c) 8.3×10^{11} Pa
(d) 8.3×10^{15} Pa
(e) None of the above

16. The Fermi momentum of a degenerate Fermi gas is given by $p_F = \hbar(3\pi^2 n)^{1/3}$, where $\hbar = \dfrac{h}{2\pi}$. For neutrons in a compact stellar core with number density $n = 5 \times 10^{30}\,\mathrm{m}^{-3}$, calculate the order of magnitude of the Fermi momentum. The value of p_F is closest to:
(a) $4.8 \times 10^{-26}\,\mathrm{kg\,m\,s^{-1}}$
(b) $4.8 \times 10^{-24}\,\mathrm{kg\,m\,s^{-1}}$
(c) $4.8 \times 10^{-20}\,\mathrm{kg\,m\,s^{-1}}$
(d) $4.8 \times 10^{-22}\,\mathrm{kg\,m\,s^{-1}}$
(e) None of the above

17. The average kinetic energy per particle in a degenerate Fermi gas at zero temperature is $\langle E \rangle = \frac{3}{5}E_F$. If the Fermi energy of electrons in a white dwarf is $E_F = 8.0 \times 10^{-18}$ J, the average kinetic energy per electron is:
(a) 15 eV
(b) 20 eV
(c) 30 eV
(d) 40 eV
(e) None of the above

18. The Fermi energy of a non-relativistic electron gas is $E_F = \frac{p_F^2}{2m_e}$. If $p_F = 1.5 \times 10^{-24}\,\mathrm{kg\,m\,s^{-1}}$, the value of E_F is approximately:
(a) 1.23×10^{-20} J
(b) 1.23×10^{-18} J
(c) 1.23×10^{-16} J
(d) 1.23×10^{-14} J
(e) None of the above

19. The Fermi energy of a non-relativistic neutron gas is $E_F = \frac{p_F^2}{2m_n}$. Using $p_F = 1.5 \times 10^{-24}\,\mathrm{kg\,m\,s^{-1}}$ estimate the neutron Fermi energy. The value of E_F is closest to:

(a) $0.7 \times 10^{-24}\,\mathrm{J}$
(b) $0.7 \times 10^{-22}\,\mathrm{J}$
(c) $0.7 \times 10^{-21}\,\mathrm{J}$
(d) $0.7 \times 10^{-19}\,\mathrm{J}$
(e) None of the above

20. Using $E_F = \frac{p_F^2}{2m_e}$, with $p_F = 2.5 \times 10^{-24}\,\mathrm{kg\,m\,s^{-1}}$, the electron Fermi energy is closest to.

(a) 1 eV
(b) 10 eV
(c) 20 eV
(d) 100 eV
(e) None of the above

21. Neutrons in a neutron star can be treated as a degenerate Fermi gas with energy–momentum relation, $E = \hbar k_F v_F$ where $v_F = 3.0 \times 10^7$ m/s. If the neutron number density is $n = 1.6 \times 10^{44}$ m^{-3} and the Fermi wave vector is given by $k_F = (3\pi^2 n)^{1/3}$. the energy of the neutron is closest to

(a) 18 MeV
(b) 33 MeV
(c) 54 MeV
(d) 87 MeV
(e) None of the above

22. The quark structure of the proton is

(a) udd
(b) uud
(c) usd
(d) uss
(e) None of the above

23. Which of the following particles is its own antiparticle?

(a) Electron
(b) Photon
(c) Proton
(d) Neutrino
(e) None of the above

24. The nuclear radius is given by the empirical relation $R = R_0 A^{1/3}$, where $R_0 = 1.3\,\mathrm{fm}$ and A is the mass number. Assuming the nucleus to be spherical, its volume is $V = \frac{4}{3}\pi R^3$. Estimate the ratio of nuclear densities of ^{208}Pb and ^{27}Al. (Take $1\,\mathrm{fm} = 10^{-15}$ m). The ratio $\rho_{\mathrm{Pb}}/\rho_{\mathrm{Al}}$ is closest to:

(a) 0.5
(b) 1
(c) 5
(d) 10
(e) None of the above

25. The binding energy of a nucleus is given by $B = [Zm_p + (A-Z)m_n - M(A,Z)]\,c^2$, where m_p and m_n are proton and neutron masses. The binding energy per nucleon is $\frac{B}{A}$. Given the atomic mass of ^{4}He is 4.002603 amu, $m_p = 1.007825$ amu, $m_n = 1.008665$ amu, estimate the binding energy per nucleon. ($1\,\mathrm{amu} = 931.5\,\mathrm{MeV/c^2}$) The value is closest to:

(a) 2 MeV
(b) 5 MeV
(c) 7 MeV
(d) 28 MeV
(e) None of the above